ML: MULTIPLE CHOICE QUESTIONS - 2

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**1.** What is the main goal of Simple Linear Regression? Answer:B
A. Classify data into categories
B. Predict a continuous outcome using one predictor
C. Reduce dimensionality
D. Cluster data

**2.** The equation of Simple Linear Regression is: Answer:B
A. $y = mx^2 + c$
B. $y = mx + c$
C. $y = \log(x)$
D. $y = e^x$

**3.** In $y = mx + c$, what does 'm' represent? Answer:C
A. Intercept
B. Error term
C. Slope
D. Mean

**4.** What does the intercept represent? Answer:A
A. Value of y when x = 0
B. Value of x when y = 0
C. Maximum value
D. Variance

**5.** Which method is commonly used to estimate parameters? Answer:B
A. Gradient Descent
B. Least Squares
C. K-Means
D. PCA

**6.** Multiple Linear Regression involves: Answer:B
A. One predictor variable
B. Two or more predictor variables

C. Only categorical variables
D. No variables

**7.** General form of Multiple Linear Regression: Answer:B
A. $y = mx + c$
B. $y = m_1x_1 + m_2x_2 + ... + c$
C. $y = x^2$
D. $y = \log(x)$

**8.** Multi-collinearity refers to: Answer:A
A. High correlation between predictors
B. High correlation between output variables
C. Low variance
D. Non-linear relationship

**9.** Which metric evaluates model fit? Answer:B
A. Accuracy
B. R-squared
C. Precision
D. Recall

**10.** Adjusted $R^2$ is used to: Answer:B
A. Increase model complexity
B. Penalize unnecessary variables
C. Ignore variables
D. Measure error

**11.** Polynomial Regression is used when: Answer:B
A. Data is linear
B. Data has a non-linear relationship
C. Data is categorical
D. Data is missing

**12.** A polynomial model of degree 2 looks like: Answer:B
A. $y = mx + c$
B. $y = ax^2 + bx + c$
C. $y = \log(x)$

D. $y = e^x$

**13.** Increasing polynomial degree may lead to: Answer:B
A. Underfitting
B. Overfitting
C. No change
D. Less variance

**14.** Polynomial regression is still: Answer:B
A. Non-linear model
B. Linear model in parameters
C. Classification model
D. Clustering model

**15.** Which technique helps avoid overfitting? Answer:B
A. Increasing features
B. Regularization
C. Ignoring data
D. Deleting model

**16.** Logistic Regression is used for: Answer:B
A. Regression problems
B. Classification problems
C. Clustering
D. Reinforcement learning

**17.** The output of Logistic Regression is: Answer:B
A. Continuous value
B. Probability between 0 and 1
C. Negative values only
D. Integers only

**18.** Which function is used in Logistic Regression? Answer:B
A. Linear function
B. Sigmoid function
C. ReLU
D. Softmax only

**19.** Sigmoid function formula is: Answer:A

A. $1 / (1 + e^{-x})$

B. $x^2$

C. log(x)

D. x + c

**20.** Decision boundary in Logistic Regression is typically: Answer:B

A. Random

B. Fixed threshold (e.g., 0.5)

C. Infinite

D. Zero only

**21.** Which regression technique is best for binary classification? Answer:B

A. Linear Regression

B. Logistic Regression

C. Polynomial Regression

D. Ridge Regression

**22.** Overfitting occurs when: Answer:B

A. Model is too simple

B. Model fits noise in data

C. Model has low accuracy

D. Model has no variables

**23.** Underfitting occurs when: Answer:B

A. Model is too complex

B. Model is too simple

C. Model fits perfectly

D. Model has high variance

**24.** Which term represents the error between predicted and actual values?
Answer:A

A. Residual

B. Gradient

C. Feature

D. Bias

**25.** Which algorithm uses a cost function like log loss? Answer:B

A. Linear Regression
B. Logistic Regression
C. Polynomial Regression
D. KNN